S2 Technical Documentation

Supplementary to:

Accelerometer Data Harmonization and Feature Engineering for Gait Analytics

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GitHub Repository

All code as well as the up-to-date technical documentation can be found in the accompanying GitHub Repository

ReichenbachLabHHN/GaitDWH: Accelerometer Data Harmonization and Feature Engineering for Gait Analytics [https://github.com/ReichenbachLabHHN/GaitDWH]

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Datasets

All data used in this study were used in accordance with the datasets' licensing terms.

Data Access

Human Gait Database

The Human Gait Database (HuGaDB) is publicly available on Kaggle at https://www.kaggle.com/datasets/romanchereshnev/hugadb-human-gait-database [1]. The data files are © the original authors. There is an original publication describing the dataset [2].

McGill University Gait Database

The McGill University Gait Database (McGill) is publicly available at https://www.cs.mcgill.ca/~jfrank8/data/gait-dataset.html [3]. A PhD-thesis is mentioning the data [4].

Movement Analysis in Real-world Environments using Accelerometers Gait Database

The release agreement and terms of usage for the Movement Analysis in Real-world Environments using Accelerometers Gait Database (MAREA) are available at https://wiki.hh.se/caisr/index.php/Gait_database [5]. There is an original publication describing the dataset [6].

Vajdi Gait Database

The Vajdi Gait Database (Vajdi) is publicly available on Kaggle at https://www.kaggle.com/datasets/drdataboston/93-human-gait-database/data [7]. The data files are © GPL 2. There is an original publication describing the dataset [8]. Note: While the paper talks about 93 participants, the actual data only contains 92 (participant 73 is missing).

Osaka University Institute of Scientific and Industrial Research Gait Database

The Osaka University Institute of Scientific and Industrial Research Gait Database (OU-ISIR) consists of several datasets, two of which have been used in this study. Note: There are small discrepancies between the gender and number of participants stated in the publications and the actual data.

OU-ISIR-Inertial Sensor Dataset

The release agreement and download link for the OU-ISIR-Inertial Sensor Dataset (OU-ISIR-ISD) is available at http://www.am.sanken.osaka-u.ac.jp/BiometricDB/InertialGait.html [9]. There is an original publication describing the dataset [10].

OU-ISIR-Similar Action Inertial Dataset

The release agreement and download link for the OU-ISIR-Similar Action Sensor Dataset (OU-ISIR-SAID) is available at http://www.am.sanken.osaka-u.ac.jp/BiometricDB/SimilarActionsInertialDB.html [11]. There is an original publication describing the dataset [12].

Database of Human Gait Performance

The Database of Human Gait Performance (DBHuGa) dataset is publicly available at:

https://doi.org/10.6084/m9.figshare.c.4892463 [13] There is an original publication describing the dataset [14].

Data Dictionary

Fact Tables

Table 1. Feature_Accelerometer_Gait_Cycle (fact table).

Column Name	Data Type	Description	Constraints / Notes
feature_accelerometer_gait_cycle_id	Integer	Unique identifier	Primary Key
Cycle	Text	List of all jerk values for this gait cycle	
minimum_index	Integer	Index of the minimum (in cycle list)	
dominant_peak_index	Integer	Index of the dominant peak (in cycle list)	
non_dominant_peak_index	Integer	Index of the non-dominant peak (in cycle list)	
segment_number	Integer	Segment number within the recording	
cycle_number	Integer	Cycle number within the segment	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id and ground_id*

segment_number*: In some recordings, several different activities were performed (e.g., first a walking sequence, then running and then walking again). In this case the recording would have two walking segments.*

Table 2. Feature_Fourier_Coefficients (fact table).

Column Name	Data Type	Description	Constraints / Notes
feature_fourier_coefficients_id	Integer	Unique identifier	Primary Key
fourier_coefficients	Text	List of all fourier coefficient values	
length_signal	Integer	Length of original jerk verctor	
frequency_peaks	Text	List of frequencies, where there is a peak	
frequency_peaks_values	Text	Amplitude of the frequency peaks	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id and ground_id*

Table 3. Feature_Mean_Accelerometer_Gait_Cycle (fact table).

Column Name	Data Type	Description	Constraints / Notes
feature_mean_accelerometer_gait_cycle_id	Integer	Unique identifier	Primary Key
value_X	Real	The individual gait cycle values at index X (0 to 199)	
amount_gait_cycles	Integer	Amount of gait cycles used for mean calculation	

This table contains extracted mean accelerometer gait cycles. Each entry represents the mean gait cycle for one recording segment of a person at a specific sensor location. This is done only for walking activity. Since the mean is calculated from the uniform gait cycles, there are 200 cycle values for each cycle.

***foreign keys** in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id and ground_id*

Table 4. Feature_Statistical-Measure_X (fact table).

Column Name	Data Type	Description	Constraints / Notes
feature_statistical_ measure_X_id	Integer	Unique identifier	Primary Key
m_dp_value	Real	Mean value of dominant peaks	
sd_dp_value	Real	Standard deviation of dominant peaks	
r_m_dnp_value	Real	Ratio of mean values of dominant and non-dominant peaks	
r_sd_dnp_value	Real	Ratio of standard deviations of dominant and non-dominant peaks	
m_sd_ds_value	Real	Mean standard deviation of dominant steps	
r_m_sd_dns_value	Real	Ratio of mean standard deviations of dominant and non-dominant steps	
sd_sd_ds_value	Real	Standard deviation of standard deviations of dominant steps	
r_sd_sd_dns_value	Real	Ratio of standard deviations of standard deviations of dominant and nondominant steps	
r_m_sl_value	Real	Ratio of means of dominant and non-dominant step length	
r_sd_sl_value	Real	Ratio of standard deviations of dominant and non-dominant step length	

Statistical Measures calculated from a windowed subsection of the recording. For a robust estimate of the feature, each one is calculated over X gait cycles X being [10, 20, 30, 40, 50].

For a reasoning behind the choice of features please refer to the publication.

***foreign keys** in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id and ground_id*

***Abbreviations:** m_dp: mean value of dominant peaks, sd_dp: standard deviation of dominant peaks, r_m_dnp = ratio of mean values of dominant and non-dominant peaks, r_sd_dnp = ratio of standard deviations of dominant and non-dominant peaks, m_sd_ds: mean standard deviation of dominant steps, r_m_sd_dns: ratio of mean standard deviations of dominant and non-dominant steps, sd_sd_ds: standard deviation of standard deviations of dominant steps, r_sd_sd_dns: ratio of standard deviations of standard deviations of dominant and nondominant steps, r_m_sl: ratio of means of dominant and non-dominant step length, r_sd_sl: ratio of standard deviations of dominant and non-dominant step length,*

Table 5. Feature_Uniform_Accelerometer_Gait_Cycle (fact table).

Column Name	Data Type	Description	Constraints / Notes
feature_uniform_accelerometer_gait_cycle_id	Integer	Unique identifier	Primary Key
value_X	Real	The individual gait cycle values at index X (0 to 199)	
minimum_index	Integer	Index of the minimum (found at value_index above)	
dominant_peak_index	Integer	Index of the dominant peak (found at value_index above)	
non_dominant_peak_index	Integer	Index of the non-dominant peak (found at value_index above)	
correlation_to_mean_recording	Real	Correlation of this cycle with the mean cycle of the same recording	
usable_flag	Boolean	True if correlation_to_mean_recording ≥0.7	
correlation_to_mean_person	Real	Correlation of this cycle with the mean cycles of the same person	
correlation_to_mean_dataset	Real	Correlation of this cycle with the mean cycle of the same dataset	
segment_number	Integer	Segment number within the recording	
cycle_number	Integer	Cycle number within the segment	

This table contains extracted uniform accelerometer gait cycles. Uniform means, the gait cycle is percentile normalized (95th percentile) and resampled to 200 values each.

***foreign keys** in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id and ground_id*

***segment_number**: In some recordings, several different activities were performed (e.g., first a walking sequence, then running and then walking again). In this case the ecoridng would have two walking segments.*

Table 6. Feature_Y_Normalized_Accelerometer_Gait_Cycle (fact table).

Column Name	Data Type	Description	Constraints / Notes
feature_accelerometer_gait_cycle_id	Integer	Unique identifier	Primary Key
cycle	Text	List of all jerk values for this gait cycle	
minimum_index	Integer	Index of the minimum (in cycle list)	
dominant_peak_index	Integer	Index of the dominant peak (in cycle list)	
non_dominant_peak_index	Integer	Index of the non-dominant peak (in cycle list)	
segment_number	Integer	Segment number within the recording	
cycle_number	Integer	Cycle number within the segment	

This table contains y-normalized (95th percentile normalized) accelerometer gait cycles.

***foreign keys** in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id and ground_id*

***segment_number**: In some recordings, several different activities were performed (e.g. first a walking sequence, then running and then walking again). In this case the recording would have two walking segments.*

Table 7. Pre_Accelerometer (fact table).

Column Name	Data Type	Description	Constraints / Notes
pre_accelerometer_id	Integer	Unique identifier	Primary Key
x	Real	Pre-processed accelerometer x-axis value	
y	Real	Pre-processed accelerometer y-axis value	
z	Real	Pre-processed accelerometer z-axis value	
time	Integer	Timestamp of the sample	
segment_number	Integer	Segment number within the recording	

This table contains the pre-processed accelerometer samples. Preprocessing consists of rotating and upscaling the data as well as segmenting the recordings into individual activities.

***foreign keys** in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id, step_event_id and ground_id*

***segment_number**: In some recordings, several different activities were performed (e.g., first a walking sequence, then running and then walking again). In this case the recording would have two walking segments.*

Table 8. Raw_Accelerometer (fact table).

Column Name	Data Type	Description	Constraints / Notes
raw_accelerometer_id	Integer	Unique identifier	Primary Key
x	Real	Raw accelerometer x-axis value	
y	Real	Raw accelerometer y-axis value	
z	Real	Raw accelerometer z-axis value	
time	Integer	Timestamp of the sample	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id, world_coordinate_system_id, step_event_id and ground_id*

Table 9. Raw_EMG (fact table).

Column Name	Data Type	Description	Constraints / Notes
raw_emg_id	Integer	Unique identifier	Primary Key
r	Real	Raw value of the right EMG	
l	Real	Raw value of the left EMG	
time	Integer	Timestamp of the sample	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id, step_event_id and ground_id*

Table 10. Raw Gyroscope (fact table).

Column Name	Data Type	Description	Constraints / Notes
raw_gyroscope_id	Integer	Unique identifier	Primary Key
x	Real	Raw gyroscope x-axis value	
y	Real	Raw gyroscope y-axis value	
z	Real	Raw gyroscope z-axis value	
time	Integer	Timestamp of the sample	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id, world_coordinate_system_id, step_event_id and ground_id*

Table 11. Raw_Jerk (fact table).

Column Name	Data Type	Description	Constraints / Notes
raw_jerk_id	Integer	Unique identifier	Primary Key
jerk_value	Real	Raw jerk value, calculated from the accelerometer data	
time	Integer	Timestamp of the sample	
segment_number	Integer	Segment number within the recording	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id, step_event_id and ground_id*

segment_number*: In some recordings, several different activities were performed (e.g. first a walking sequence, then running and then walking again). In this case the recording would have two walking segments.*

Table 12. Raw_Magnetometer (fact table).

Column Name	Data Type	Description	Constraints / Notes
raw_magnetometer_id	Integer	Unique identifier	Primary Key
x	Real	Raw magnetometer x-axis value	
y	Real	Raw magnetometer y-axis value	
z	Real	Raw magnetometer z-axis value	
time	Integer	Timestamp of the sample	

foreign keys *in this table: recording_id, person_id, sensor_id, activity_id, sensor_location_id, world_coordinate_system_id, step_event_id and ground_id*

Dimension Tables

Table 13: Activity Table (dimension table).

Column Name	Data Type	Description	Notes
activity_id	Integer	Unique identifier	Primary Key
activity_type_id	Integer	Rough categorization of the activity	
activity_type_title	Text	Corresponding title to an activity_id	
activity_name_id	Integer	Specification of activity	
activity_name_title	Text	Corresponding title to an activity_name_id	

Table 14. activity_type_id and corresponding activity_type_title.

0: Active Walking	1: Active Other	2: Passive	3: Invalid	4: Mixed

Table 15. activity_name_id and corresponding activity_name_title.

0: Walking	1: Running	2: Going Slope Up	3: Going Slope Down
4: Going Stairs Up	5: Going Stairs Down	6: Sitting	7: Sitting Down
8: Standing Up	9: Standing	10: Bicycling	11: Up By Elevator
12: Down By Elevator	13: Sitting In Car	14: Walking & Running	15: Invalid
16: Mixed			

Table 16. activity_id specification

activity_id	activity_type_title	activity_name_title
1	Active Walking	Walking
2	Active Walking	Running
3	Active Walking	Going slope up
4	Active Walking	Going slope down
5	Active Walking	Going stairs up
6	Active Walking	Going stairs down
7	Passive	Sitting
8	Active Other	Sitting down
9	Active Other	Sitting up
10	Passive	Standing
11	Active Other	Bicycling
12	Passive	Up by elevator
13	Passive	Down by elevator
14	Passive	Sitting in car
15	Active Walking	Walking & Running
16	Invalid	Invalid
17	Mixed	Mixed

each activity_id corresponds to a specific pair of activity_type and activity_name as indicated in this table

Table 17. Ground (dimension table).

Column Name	Data Type	Description	Constraints / Notes
ground_id	Integer	Unique identifier	Primary Key
outside_id	Integer	Specifies if data was recorded outside	
outside_title	Text	Corresponding title to an outside_id	
even_slope_stairs_id	Integer	Rough categorization of ground characteristics	
even_slope_stairs_title	Text	Corresponding title to an even_slope_stairs_id	
treadmill_id	Integer	Specifies if data was recorded on a treadmill	
treadmill_title	Text	Corresponding title to a treadmill_id	
surface_id	Integer	Specification of ground characteristics	
surface_title	Text	Corresponding title to a surface_id	

This table contains information about the ground the data was recorded on.

Table 18. outside_id and corresponding outside_title.

0: Inside	1: Outside	2: Invalid

Table 19. even_slope_stairs_id and corresponding even_slope_stairs_title.

0: Even	1: Slope	2: Stairs	3: Others
4: Various	5: Invalid		

Table 20. treadmill_id and corresponding treadmill_title.

0: Natural	1: Treadmill	2: Invalid

Table 21. surface_id and corresponding surface_title.

0: Standard Indoor Surface	1: Paved	2: Cement	3: Grass
4: Uneven Stone Brick	5: Various	6: Invalid	

Table 22. Person (dimension table).

Column Name	Data Type	Description	Constraints / Notes
person_id	Integer	Unique identifier	Primary Key
height	Real	Height of the person in cm	
weight	Real	Weight of the person in kg	
age	Integer	Age of the person in years	
sex_id	Integer	Specifies the persons sex	
sex_title	Text	Corresponding title to a sex_id	
smoker_id	Integer	Specifies if a person is a smoker	
smoker_title	Text	Corresponding title to a smoker_id	
fitness_level	Integer	Time person exercises per week in minutes	

Table 23. sex_id and corresponding sex_title.

0: Male	1: Female

Table 24. smoker_id and corresponding smoker_title.

0: Non-Smoker	1: Smoker

Table 25. Recording (dimension table).

Column Name	Data Type	Description	Notes
recording_id	Integer	Unique identifier	Primary Key
year	Integer	Year the recording was taken in	
month	Integer	Month the recording was taken in	
day	Integer	Day the recording was taken at	
starting_time	Text	Starting time of the recording in form ‘HH:MM’	
duration	Integer	Duration of recording in milliseconds	
dataset_id	Integer	Specifies the dataset the recording is from	
dataset_title	Text	Corresponding title to a dataset_id	
repetition	Integer	Counter for the repetition of the same trial for a person	
sample_frequency	Real	The recording’s sample frequency	

Table 26. dataset_id and corresponding dataset_title.

0: Human Gait Database	1: McGill University Gait Database
2: MAREA Gait Database	3: Vajdi Gait Database
4: OU-ISIR Inertial Sensor Dataset 1	5: OU-ISIR Inertial Sensor Dataset 2
6: OU-ISIR Similar Action Inertial Dataset	7: A Database of Human Gait Performance

Table 27. Sensor (dimension_table).

Column Name	Data Type	Description	Notes
sensor_id	Integer	Unique identifier	Primary Key
modality_id	Integer	Specifies the sensor's modality type	
modality_title	Text	Corresponding title to a modality_id	
model	Text	Specifies model of sensor or device it is built into	
sensor_type_id	Integer	Specifies sensor's modality	
sensor_type__title	Text	Corresponding title to a sensor_type_id	
internal_coordinate_system_id	Integer	Specifies internal coordinate system (x-, y- and z-direction of sensor)	
internal_coordinate_system_title	Text	Corresponding title to an internal_coordinate_system_id	
rotation_to_standard_coordinate_system	Text	Rotation Matrix to rotate sensor's internal coordinate system to globally chosen 'world coordinate system' (x right, y superior, z posterior)	

Table 28. modality_id and corresponding modality_title.

0: Accelerometer 1: Gyroscope 2: EMG 3: Magnetometer

Table 29. sensor_type_id and corresponding sensor_type_title.

0: Inertial Measurement Unit 1: Smartphone Sensor

Table 30. internal_coordinate_system_id and corresponding internal_coordinate_system_title.

0: x right, y superior, z anterior 1: x superior, y left, z anterior

2: x right, y anterior, z superior

Table 31. Sensor_Location (dimension table).

Column Name	Data Type	Description	Notes
sensor_location_id	Integer	Unique identifier	Primary Key
body_side_id	Integer	Specifies the body side the sensor is placed on	
body_side_title	Text	Corresponding title to a body_side_id	
body_area_id	Integer	Rough categorization of body area the sensor is located in	
body_area_title	Text	Corresponding title to a body_area_id	
position_id	Integer	Specification of body are the sensor is located in	
position_title	Text	Corresponding title to a position_id	
rotation	Text	Rotation matrix to rotate the sensor to compensate for the way it is attached to the body	

This table contains information about the sensor location. Multiple sensors can share a common sensor_location entry.

Table 32. body_side_id and corresponding body_side_title.

0: Left	1: Center	2: Right

Table 33. body_area_id and corresponding body_area_title.

0: Foot	1: Ankle	2: Shin	3: Thigh
4: Pocket	5: Waist	6: Wrist	

Table 34. position_id and corresponding position_title

0: Top Of Left Foot	1: Top Of Right Foot	2: Left Ankle
3: Right Ankle	4: Left Shin	5: Right Shin
6: Left Thigh Front	7: Right Thigh Front	8: Left Vastus Lateralis
9: Right Vastus Lateralis	10: Left Front Pocket	11: Left Back Pocket
12: Right Front Pocket	13: Center Waist Back	14: Left Waist
15: Right Waist	16: Center Waist Front	17: Left Wrist
18: Right Wrist		

Table 35. Step Event (dimension table).

Column Name	Data Type	Description	Notes
step_event_id	Integer	Unique identifier	Primary Key
ground_truth_event_id	Integer	Encodes the ground truth events	
ground_truth_event_title	Text	Corresponding title to a ground_truth_event_id	
ground_truth_order_flag	Boolean	Flag stating, if ground truth event is in correct order with previous and following event	
ground_truth_step_number	Integer	Step number, the sample belongs to in recording according to ground truth	
detected_event_id	Integer	Encodes detected gait cycle event	
detected_event_title	Text	Corresponding title to a detected_event_id	
detected_event_reliability	Real	Reliability of detected event. Deduced from correlation of the gait cycle this event belongs to with mean gait cycle of the recording.	
detected_event_reliable_flag	Boolean	States if detected event is reliable (detected event reliability ≥ 0.7)	

This table contains ground truth data provided by the data sources, as well as gait cycle events extracted from the data.

Table 36. ground_truth_event_id and corresponding ground_truth_event_title.

0: No Event	1: Toe-Off Event Left	2: Toe-Off Event Right
3: Heel-Strike Event Left	4: Heel-Strike Event Right	

Table 37. detected_event_id and corresponding detected_event_title.

0: No Event	1: Dominant Peak	2: Non-Dominant Peak	3: Minimum

Table 38. world_coordinate_system (dimension table).

Column Name	Data Type	Description	Notes
world_coordinate_system_id	Integer	Unique identifier	Primary Key
alignment_id	Integer	Specifies how sensor is aligned (combination of internal coordinate system and rotation of sensor's attachment)	
alignment_title	Text	Corresponding title to an alignment_id	
rotation_to_world_coordinate_system	Text	Rotation matrix to rotate the sensor to world coordinate system (x right, y superior, z anterior)	

Table 39. alignment_id and corresponding alignment_title.

0: x superior, y right, z posterior	1: x posterior, y right, z inferior	2: x right, y inferior, z posterior
3: x left, y inferior, z anterior	4: x posterior, y left, z inferior	5: x right, y posterior, z inferior
6: x posterior, y superior, z right	7: x left, y superior, z posterior	8: x anterior, y superior, z left
9: x superior, y left, z anterior	10: x superior, y posterior, z left	11: x anterior, y inferior, z right
12: x posterior, y inferior, z left		

This table contains information about how sensors are aligned globally, and how to rotate them to fit into a common coordinate system.

Data Mapping

Sensors

This section details the ways rotation matrices to the global coordinate system were calculated for the various data sources.

The global coordinate system is defined as **x right, y superior, z anterior**.

Calculating the global rotation matrix requires two stages:

1) The internal coordinate system of the recording sensor.
2) The rotation of the sensor due to its attachment to the body.

By rotating the sensors, we ensure that the three axes all refer to the same direction, and that their values are therefore comparable.

In our analysis we use the Euclidean norm of the jerk, which is independent from orientational influences.

Additionally, some data sources use smartphones as recording devices, which, if carried in a pocket, move around during the recording, and therefore may change their orientation. While there are ways to mitigate this effect, in this case, no such steps were taken.

Note: The datasets define left and right from the participant's perspective, whereas this study uses an observer-based reference frame. Accordingly, the datasets' left/right labels were inverted to match our convention.

Human Gait Database

The sensors used in the HuGaDB recordings are of the model MPU9250 [2]. Using the sensor's documentation [15], we can find its internal coordinate system, which is x right, y superior, z anterior (Fig 1).

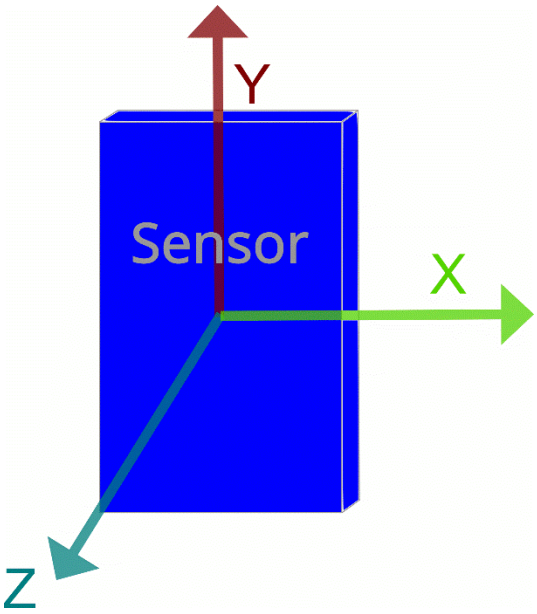


***Fig 1. Internal coordinate system of the Human Gait Database** [15].*

Using this information, we can calculate a general rotation matrix for transforming the sensor's data into the global coordinate system. Since all axes are already matching, the general rotation matrix for this sensor ends up being [1, 0, 0] [0, 1, 0] [0, 0, 1].

Note: While the MPU9250 sensor includes a magnetometer, this functionality was not used in the study. A standalone EMG sensor was used in the study, but since no further information is given, we cannot calculate rotation matrices for it.

As for the sensor orientation due to the body attachment, the HuGaDB documentation [16] contains a visualization which this information can be extracted from based on which is Fig 2.

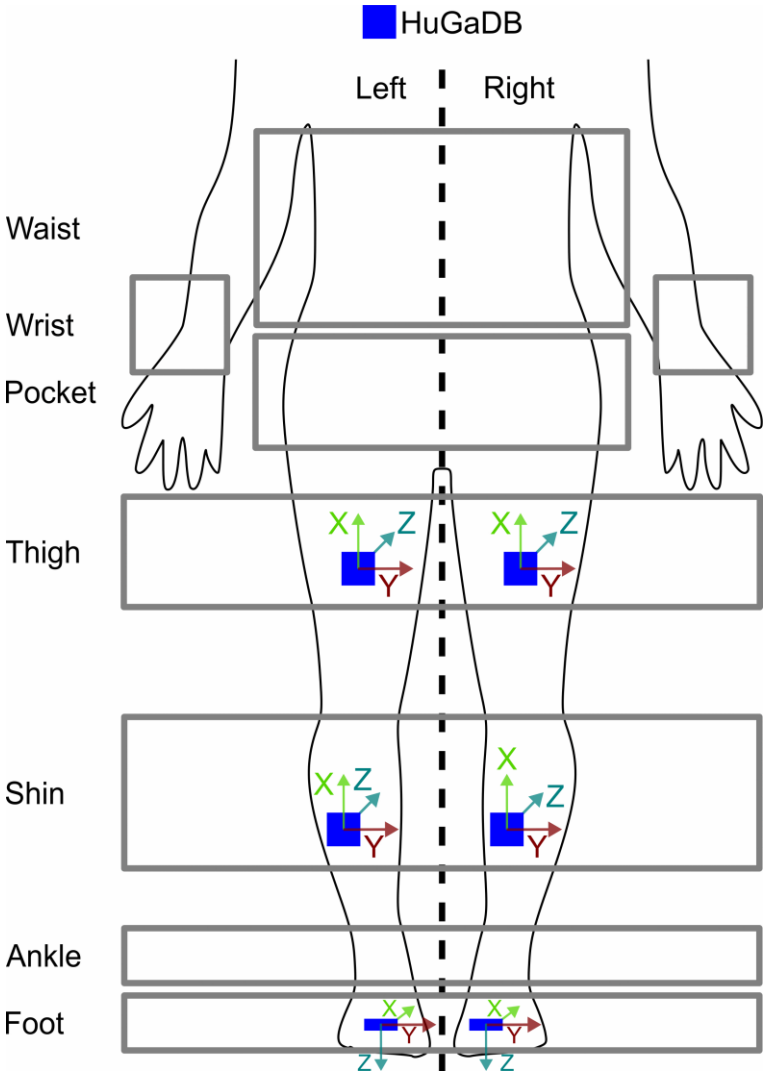


***Fig 2. Sensor position and rotation of the Human Gait Database**. Depicted is a front view of the participant.*

With this information, we can calculate a rotation matrix, which is supposed to compensate for the sensor's rotation due to its attachment to the body. Since not all sensors are attached the same way, we end up with two matrices, one for the sensors on the feet, and one for the other sensors.

In the latter case, the sensor is turned by 180 degrees around the y-axis and 90 degrees around the z-axis resulting in a rotation matrix of [0, 1, 0] [1, 0, 0] [0, 0, -1].

The feet sensors are additionally turned negative 90 degrees around the y-axis and therefore result in a rotation matrix of [0, 1, 0] [0, 0, -1] [-1, 0, 0].

To calculate the total rotation matrices to the global coordinate system, we multiply the internal sensor rotation matrix with the body attachment rotation matrix.

The results are:

Feet sensors: [0, 1, 0] [0, 0, -1] [-1, 0, 0]

Other sensors: [0, 1, 0] [1, 0, 0] [0, 0, -1]

McGill University Gait Database

The McGill database gives little information about axis orientations except the model of smartphone used for its recordings, which is the Google Nexus One. A publication from 2014 [17] provides us with its supposed internal coordinate system, which is x right, y superior, z anterior (Fig 3).

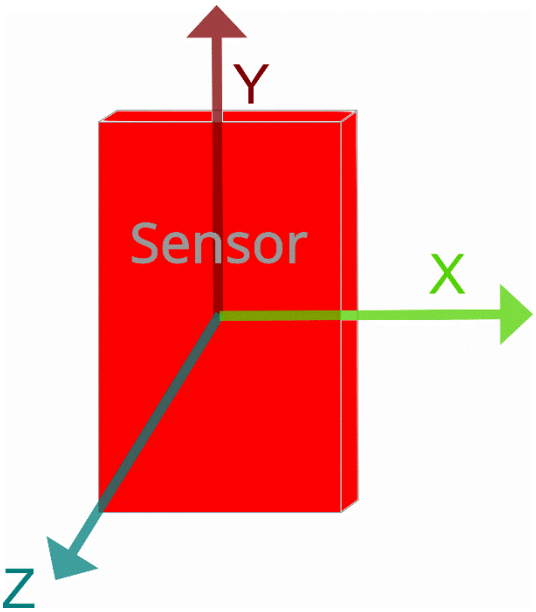


Fig 3. Internal coordinate system of the McGill University Gait Database.

Due to the lack of alternate sources, this information was assumed to be correct.

The rotation to the standard internal coordinate system therefore is [1, 0, 0] [0, 1, 0] [0, 0, 1].

The McGill database uses smartphones in three locations: left front pocket, right front pocket, and left back pocket. Since there is no information about the attachment to the body 'natural hand movements' of how one puts a phone in the pocket were assumed.

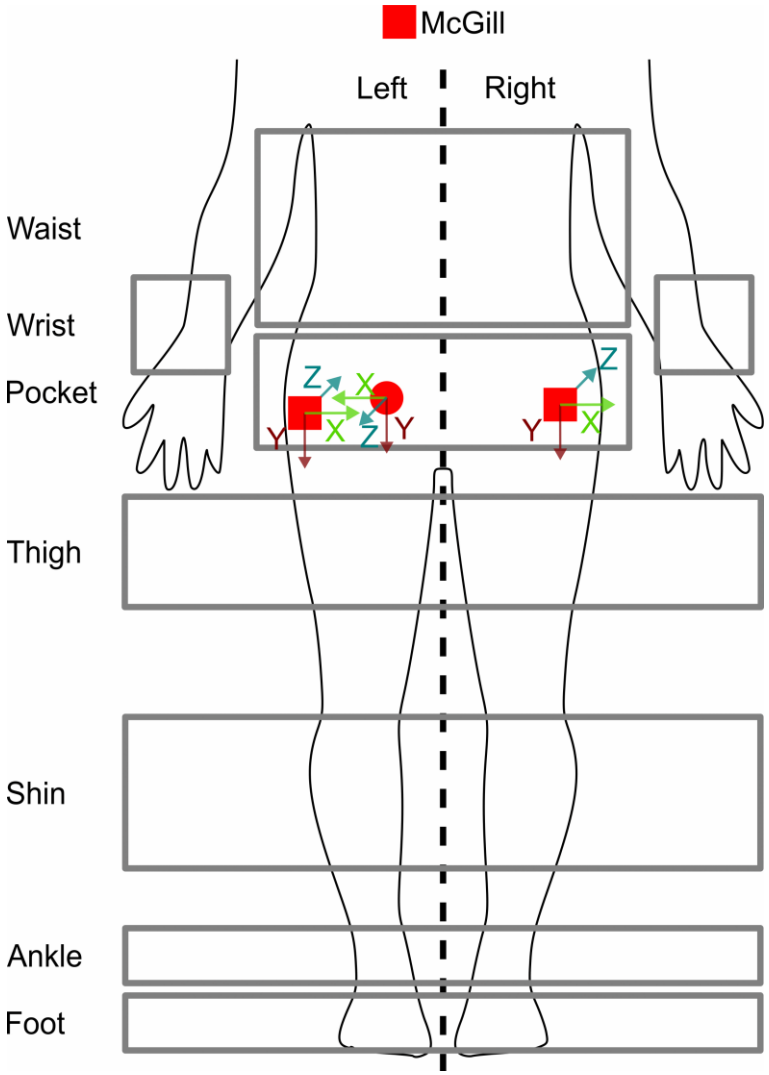


Fig 4. Sensor position and rotation of the MCGill University Gait Database.

Depicted is a front view of the participant. The circle indicates sensor placement on the back.

The left and right front pockets share the same sensor rotation, 180 degrees around the x-axis, and therefore [1, 0, 0] [0, -1, 0] [0, 0, -1].

The sensor in the left back pocket on the other hand is assumed to be turned both 180 degrees around the x- and y-axis, and therefore has a [-1, 0, 0] [0, -1, 0] [0, 0, 1] rotation (**Fehler! Verweisquelle konnte nicht gefunden werden.**).

The final results are:

Front pockets: [1, 0, 0] [0, -1, 0] [0, 0, -1]

Back pocket: [-1, 0, 0] [0, -1, 0] [0, 0, 1]

MAREA Gait Database

The MAREA dataset states in its documentation [5], that it uses Shimmer3 sensors for its recordings. Using the sensor's documentation [18], we can find its internal coordinate system, which is x right, y superior, z anterior (**Fehler! Verweisquelle konnte nicht gefunden werden.**).

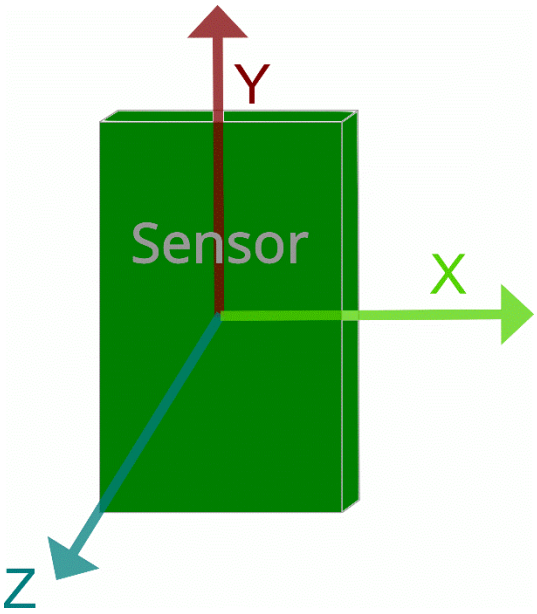


Fig 5. Internal coordinate system of the MAREA Gait Database.

With this, we can calculate the internal rotation matrix: [1, 0, 0] [0, 1, 0] [0, 0, 1]

Note: While the Shimmer3 sensor includes a gyroscope and a magnetometer, this functionality was not used in the study.

The dataset documentation also provides us with an image containing the sensor attachment axes but only showing two of the three axes. Since we have the internal sensor axis configurations, we can deduce the missing third axis, except for the left ankle sensor (**Fehler! Verweisquelle konnte nicht gefunden werden.**). For the left ankle sensor, the documentation states that its attachment is arbitrary.

The left ankle sensor was assumed to be rotated 180 degrees around the y-axis in relation to the right ankle sensor.

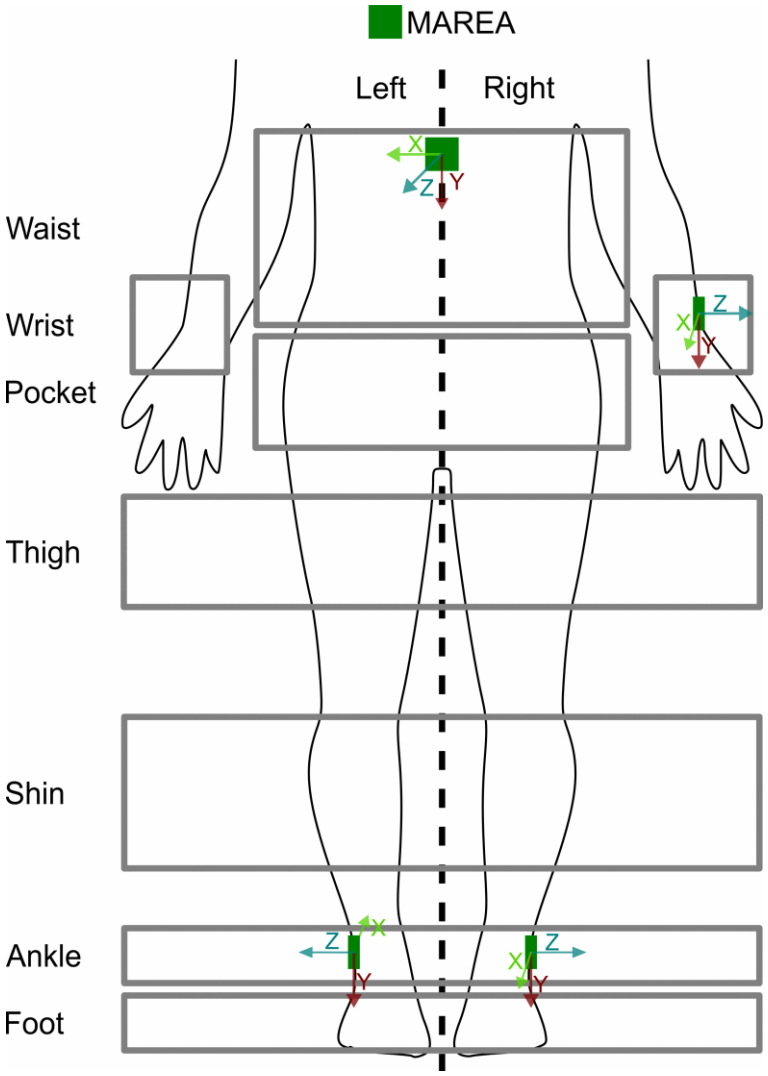


Fig 6. Sensor position and rotation of the MAREA Gait Database. Depicted is a front view of the participant.

Since right ankle and right wrist sensors are attached with the same orientation, we end up with 3 attachment rotation matrices:

Right ankle/right wrist: [0, 0, 1] [0, -1, 0] [1, 0, 0]

Left ankle: [0, 0, -1] [0, -1, 0] [-1, 0, 0]

Waist center: [-1, 0, 0] [0, -1, 0] [0, 0, 1]

The final results are:

Right ankle/right wrist: [0, 0, 1] [0, -1, 0] [1, 0, 0]

Left ankle: [0, 0, -1] [0, -1, 0] [-1, 0, 0]

Waist center: [-1, 0, 0] [0, -1, 0] [0, 0, 1]

Vajdi Gait Database

The Vajdi dataset works with two iPhone 6S smartphones. According to an in depth teardown of the phone [19], it has two separate accelerometer sensors, of which only one also records gyroscope data. Since the Vajdi dataset provides both, this is assumed to be the relevant sensor which is the InvenSense MP67B model. The teardown also states that the iPhone 6 shares the same inertial sensor with its S-variant. This is relevant, since, while we cannot find any axis orientations for the iPhone 6S, we are able to find one for the non-S-version [20] which is x right, y anterior, z superior (Fig 7).

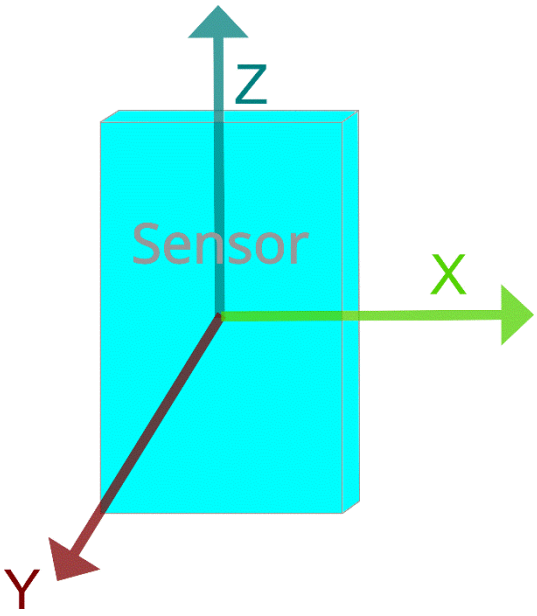


Fig 7. Internal coordinate system of the Vajdi Gait Database.

Therefore, the internal rotation matrix ends up being [1, 0, 0] [0, 0, 1] [0, 1, 0].

As for the attachment axis orientations, there is no specific information given, so the same assumption as for the McGill sensor, that the smartphone in the right pocket is turned 180 degrees around the x-axis, was made.

As for the smartphone on the waist, it is likely that the same orientation was used with the addition of turning it negative 90 degrees around the z-axis.

This results in the axes orientation of the sensors seen in Fig 8.

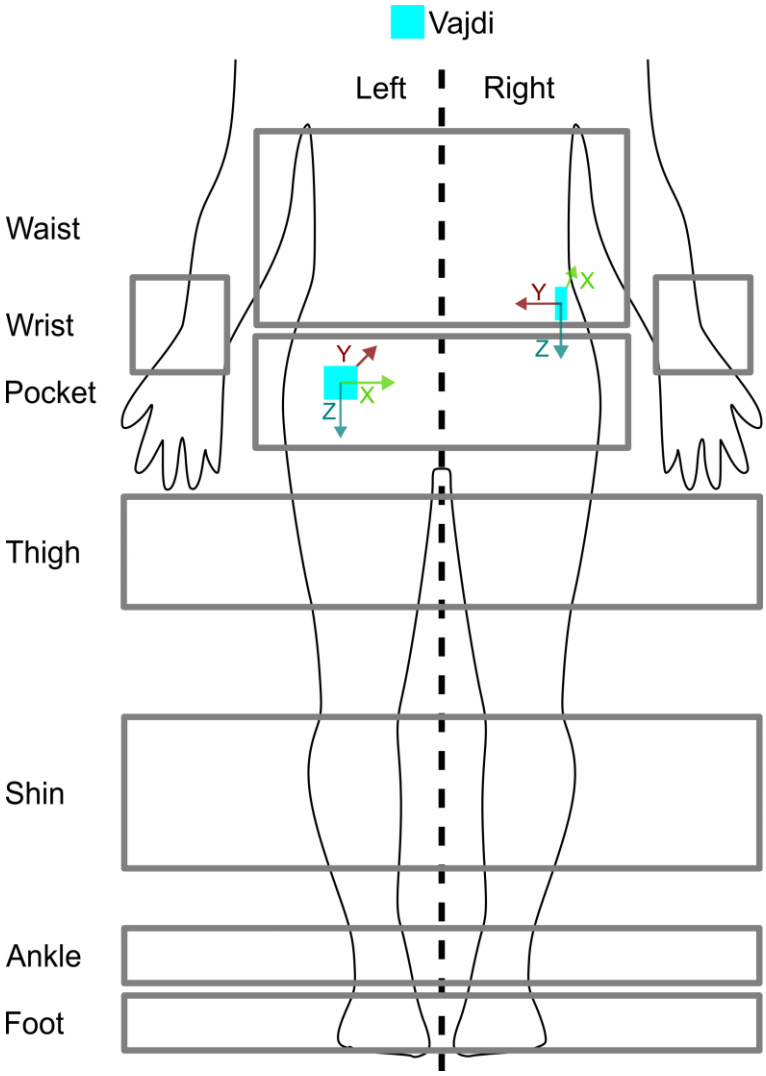


Fig 8. Sensor position and rotation of the Vajdi Gait Database. *Depicted is a front view of the participant.*

We therefore end up with two sensor attachment rotation matrices:

Left pocket: [1, 0, 0] [0, -1, 0] [0, 0, -1]

Right Waist: [0, -1, 0] [-1, 0, 0] [0, 0, -1]

The final results are:

Left pocket: [1, 0, 0] [0, 0, -1] [0, -1, 0]

Right Waist: [0, -1, 0] [0, 0, -1] [-1, 0, 0]

OU-ISIR Gait Database

The publications and documentation accompanying the OU-ISIR data [9-12] supply all necessary axis orientations information we need for its inertial sensor, which is x right, y superior, z anterior (Fig 9).

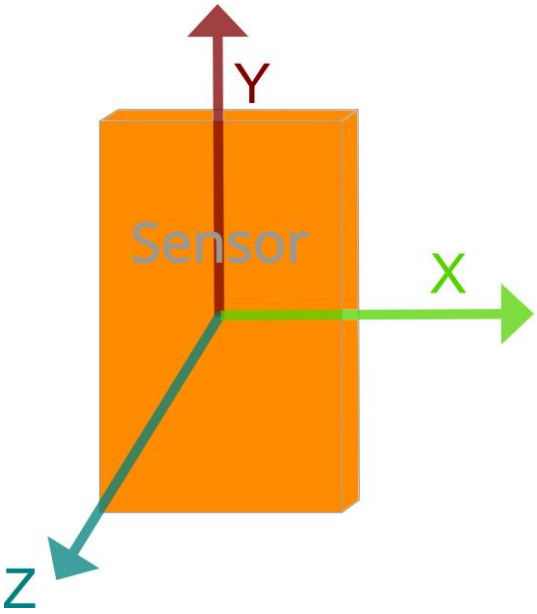


Fig 9 Internal coordinate system of the Inertial Sensor Dataset (IMUZ sensor).

We can therefore directly calculate the internal rotation matrix: [1, 0, 0] [0, 1, 0] [0, 0, 1]

The above-mentioned publications and documentation [9-12] also show how the sensors are attached to the body which results in the axis orientations seen in Fig 10.

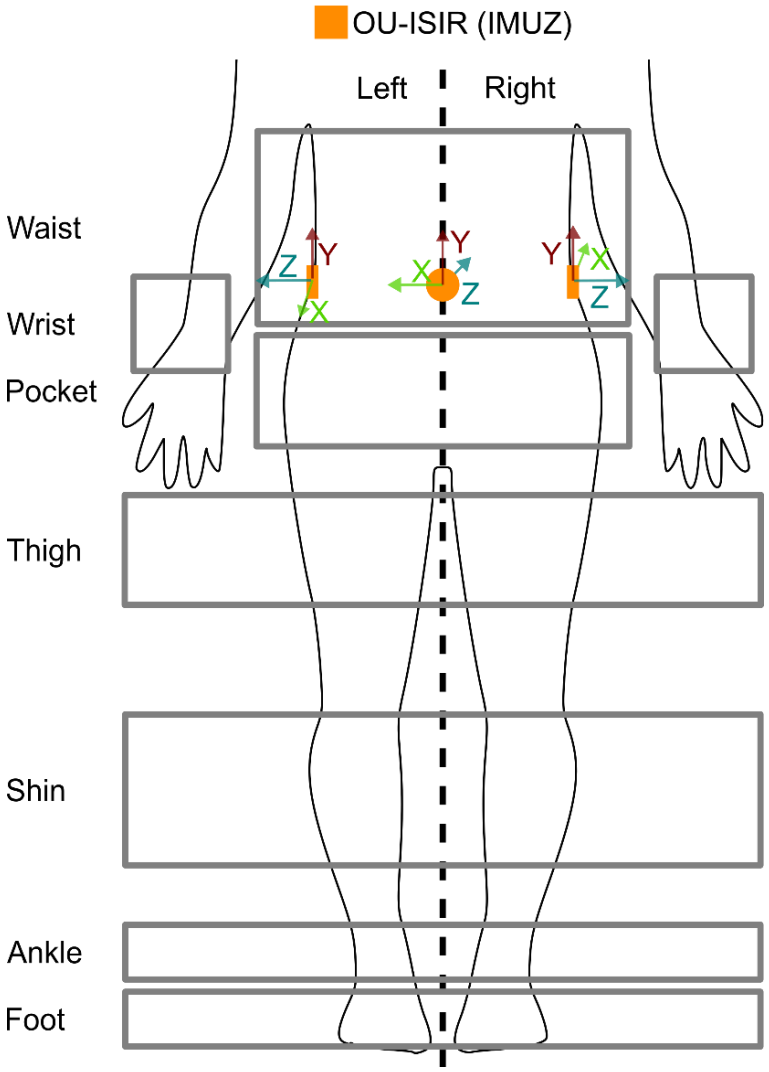


Fig 10 Sensor position and rotation of the Inertial Sensor Dataset (IMUZ sensor).

Depicted is a front view of the participant.

Waist back center: [-1, 0, 0] [0, 1, 0] [0, 0, -1]

Waist right: [0, 0, 1] [0, 1, 0] [-1, 0, 0]

Waist leftt: [0, 0, -1] [0, 1, 0] [1, 0, 0]

The final results are:

Waist back center: [-1, 0, 0] [0, 1, 0] [0, 0, -1]

Waist right: [0, 0, 1] [0, 1, 0] [-1, 0, 0]

Waist left: [0, 0, -1] [0, 1, 0] [1, 0, 0]

The OU-ISIR dataset also includes recordings of a Motorola ME860.

The Motorola ME860 is also known as Motorola Atrix 4G. For this name, there is a teardown documentation [21], which provides the name of its accelerometer, the Kionix KXTF9 MEMS Motion Sensing Accelerometer. The sensor's data sheet [22] provides its internal coordinate system.

With the help of the teardown, we can deduce that the accelerometer is turned by 90 degrees around the z-axis, and that the actual internal coordinate system of the phone is x superior, y left, z anterior (Fig 11).

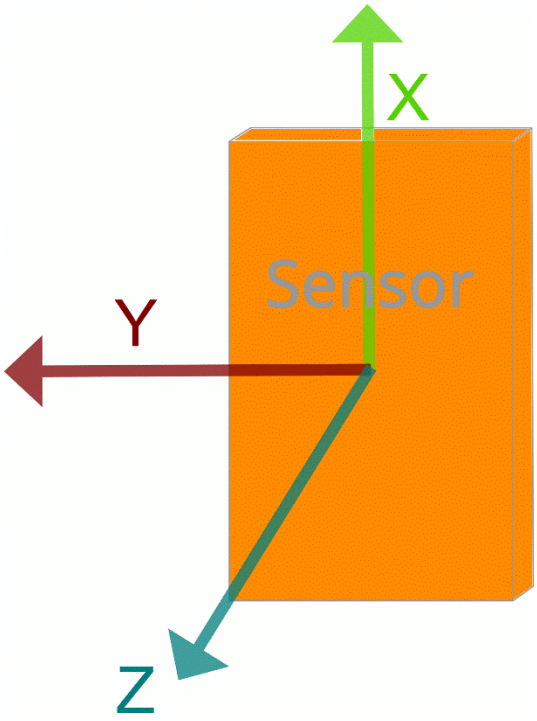


Fig 11 Internal coordinate system of the Inertial Sensor Dataset (Motorola sensor).

The internal rotation matrix therefore is [0, -1, 0][1, 0, 0][0, 0, 1].

The body attachment image in the dataset's publication and documentation [9, 10] show that the phone is turned by 180 degree around the sensor's x-axis, resulting in the axis orientations seen in Fig 12.

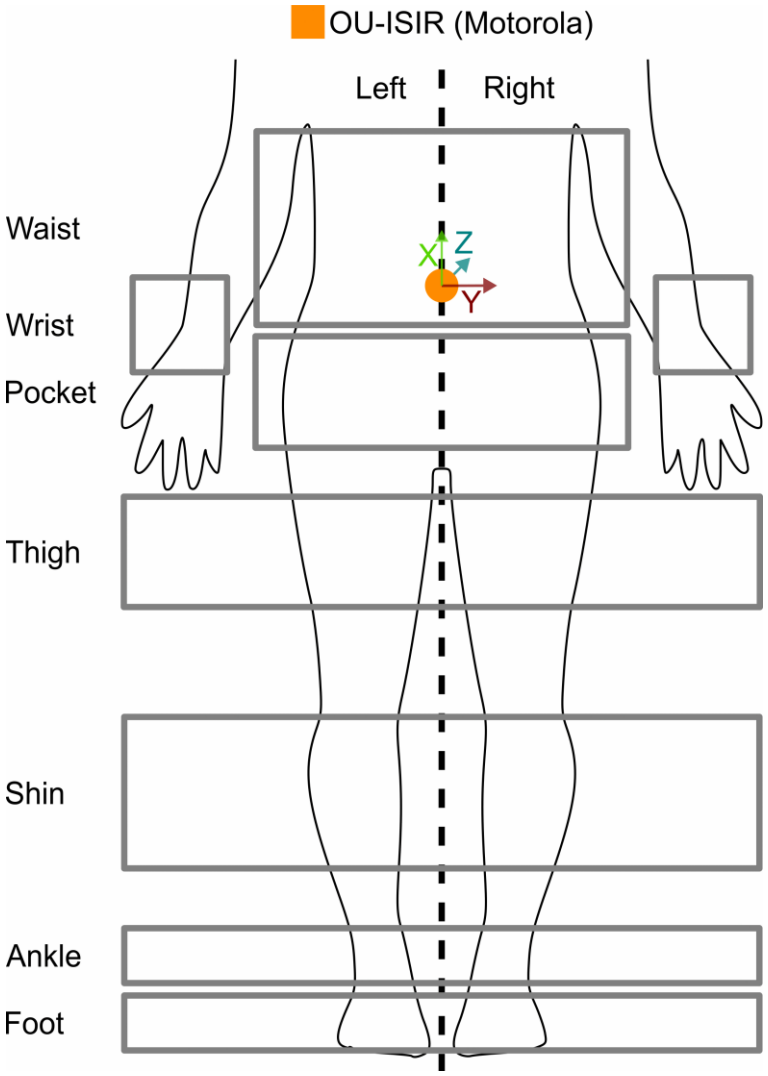


Fig 12 Sensor position and rotation of the Inertial Sensor Dataset (Motorola sensor).

Depicted is a front view of the participant.

The rotation matrix therefore is [1, 0, 0] [0, -1, 0] [0, 0, -1].

The final result is:

Smartphone sensor: [0, 1, 0] [1, 0, 0] [0, 0, -1]

Database of Human Gait Performance

The publication for this dataset [14] provided the model of sensor used, namely the XSens MTw Awina. Looking into its documentation [23], we can find its standard internal coordinate system, which is x superior, y left, z anterior (Fig 13).

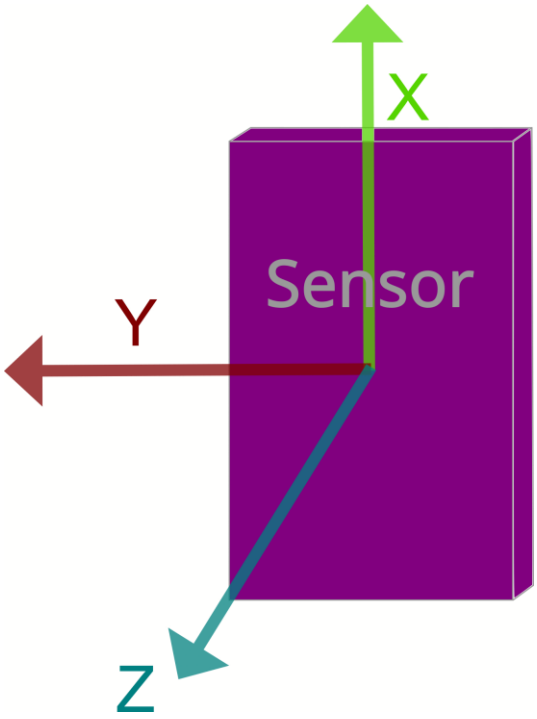


Fig 13 Internal coordinate system of the Database of Human Gait Performance.

Using this information, we can calculate the internal rotation matrix: [0, -1, 0] [1, 0, 0] [0, 0, 1]

As for the body attachment axis configurations, the provided images are inconclusive. It is assumed that the sensor is usually worn with the XSENS logo outwards and upright, resulting in the axis orientations seen in Fig 14.

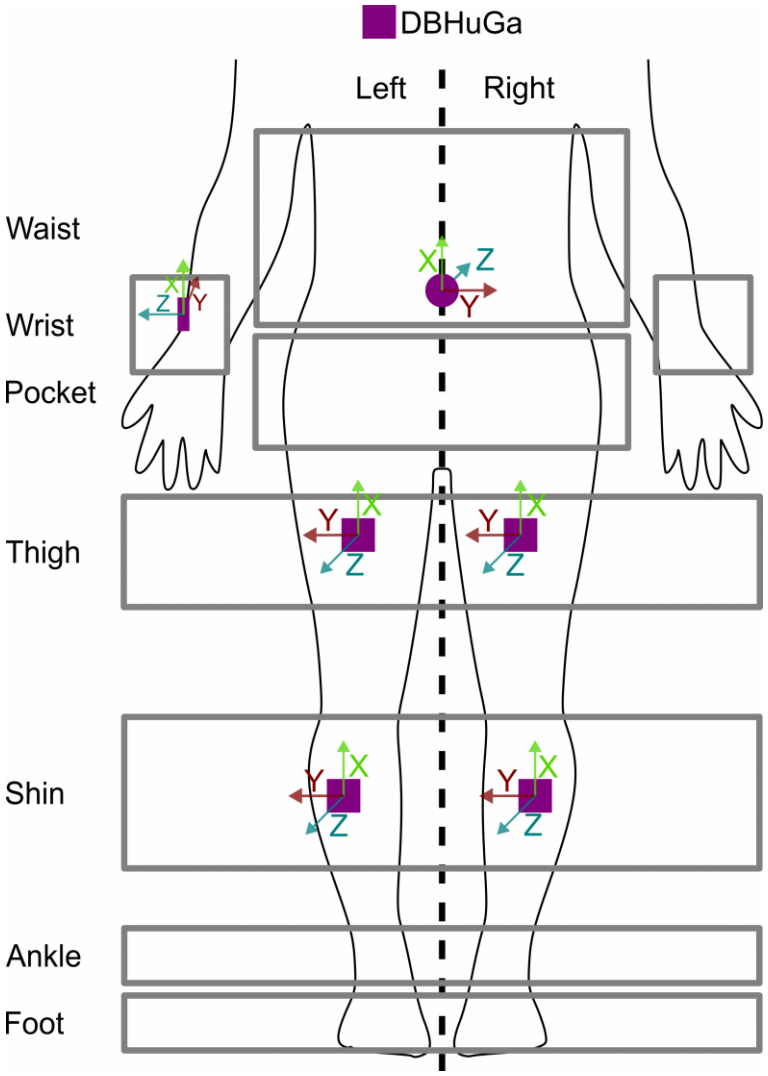


Fig 14 Sensor position and rotation of the Database of Human Gait Performance.

Depicted is a front view of the participant.

In this case, the rotation matrices are:

Thigh and shin: [1, 0, 0][0, 1, 0][0, 0, 1]

Waist back center: [1, 0, 0][0, -1, 0][0, 0, -1]

Left wrist: [1, 0, 0][0, 0, 1][0, -1, 0]

The final results are:

Thigh and shin: [0, -1, 0][1, 0, 0][0, 0, 1]

Waist back center: [0, 1, 0][1, 0, 0][0, 0, -1]

Left wrist: [0, 0, -1][1, 0, 0][0, -1, 0]

Algorithms

Extract-Transform-Load (ETL)

The data was extracted through individual scripts. If researchers acquire the same datasets our scripts can be used to transform them into our data schema. They can be found in the 'ETL' folder of the provided code. For other datasets, new scripts must be written.

Since every dataset differs in the way it saves its dimension data there is no one fits all solution for loading the meta data into the database. For our solution we created csv files for each relevant dimension (like person or sensor) in which we wrote the individual metadata for each dataset.

Those files were later used when filling the database. Researchers using this software package need to expand and alter those meta data files, depending on which datasets they use.

The python file 'all_in_one.py' in the 'ETL' folder sets up the database and fills it with the provided meta data.

To set up the database, the '.envs' file in the 'Database' folder needs to be changed according to the database settings.

We have uploaded an Excel file (database_documentation_etl_process_mapping) with a table for each dataset in the GitHub Repository. The information on how the original files were transformed into the initial dimension and fact tables of our data schema can be found there.

File Structure

To be able to use the given ETL scripts you need to organise the datasets (once obtained) as follows.

All the Data needs to be stored in the 'Data' folder which is located in the 'Database' folder. Within this folder each dataset gets their own folder.

HuGaDB

The HuGaDB folder needs to be named '00_HuGaDB'. Within this folder there needs to be a subfolder 'HuGaDB_vs'. The raw data text files should be placed within this subfolder.

McGill

The McGill folder needs to be named '01_McGill'. Within this folder there needs to be the 'SubjectInformation.csv' file as well as a subfolder called "gait-dataset". The raw data .csv files should be placed within this subfolder.

MAREA

The MAREA folder needs to be named '02_MAREA'. Within this folder there needs to be the 'GroundTruth.mat' file as well as subfolders called "Activity_Timings" and 'Subject_Data_txt_format'. Within these subfolders the Timings.mat files and the raw data text files should be placed respectively.

Vajdi

The Vajdi folder needs to be named ‘03_Vajdi’. Within this folder there needs to be a subfolder called “gait-data”. The raw data .csv files should be placed within this subfolder.

Since the meta data were added manually to a separate file (see Extract-Transform-Load (ETL)) the ‘FinalMatrix-Gait-Meta-Data’ files are not necessary. Researchers are free to use that file with a custom script of their own.

OU-ISIR

The OU-ISIR folder needs to be named ‘04_OU-ISIR’. Within this folder there need to be subfolders for each dataset belonging to this database.

OU-ISIR-ISD 1

This subfolder needs to be named ‘Inertial_Sensor_Dataset_1.’ The raw data .csv files should be placed in this folder.

OU-ISIR-ISD 2

This subfolder needs to be named ‘Inertial_Sensor_Dataset_2.’ Each sensor position gets its own subfolder as follows: ‘Android’, ‘IMUZCenter’, ‘IMUZLeft’ and ‘IMUZRight’. Note again that we define left and right opposite of the dataset. The raw data .csv files need to be placed in the corresponding sensor folder.

Similar_Action_Inertial_Dataset

This subfolder needs to be named ‘Inertial_Sensor_Dataset_2’. Each sensor position gets its own subfolder as follows: ‘Android’, ‘CenterSensor’, ‘LeftSensor’ and ‘RightSensor. Note again that we define left and right opposite of the dataset. The raw data .txt files need to be placed in the corresponding sensor folder.

Since the meta data were added manually to a separate file (see Extract-Transform-Load (ETL)) the ‘IDGenderAgelist.csv’ file is not necessary. Researchers are free to use that file with a custom script of their own.

DBHuGa

The DBHuGa folder needs to be named ‘009_A_database_for_human_gait’. Within this folder there needs to be a subfolder called “subjects”. Within this subfolder each subject gets their own numbered folder (one to thirty). Here the individual subjects’ raw data files should be placed.

Since the meta data were added manually to a separate file (see Extract-Transform-Load (ETL)) the ‘meta_data’ files are not necessary. Researchers are free to use that file with a custom script of their own.

Feature Extraction

Adjustable Parameters

The feature extraction pipeline is found in the 'extract_features.py' file in the folder 'Feature_Extraction'.

There are seven configurable parameters that control data selection, preprocessing, and feature calculation. These parameters can be individually modified via the '.envs' file in the 'Database' folder.

activities_list:

This parameter defines the activities to be included during feature extraction. The default value is '1' (corresponding to pure walking). Valid activity IDs can be referenced in the *Activity Table* in the Data Dictionary. The relevant field is activity_id, which serves as the primary key.

feature_frames_list:

This parameter specifies the number of gait cycles used when calculating statistical features. The default value is '10, 20, 30, 40, 50'. Each value represents a window size (in gait cycles).

resample_target_sample_rate:

Controls the sampling frequency in Hertz to which all accelerometer data are resampled. The default value is '128'**.** This corresponds to the highest sampling frequency available within the datasets used in this study (MAREA Gait Database).

moving_average_filter_window_size:

Defines the window size of the moving average filter applied when calculating the jerk vector from accelerometer data. The default value is '10'.

fourier_coefficient_amount:

Specifies the number of Fourier coefficients extracted from each accelerometer segment. Each recording is divided into segments according to the detected activity before coefficient extraction. The default value is '40'. Only a limited subset of coefficients is stored, meaning that the original signal cannot be fully reconstructed from the stored coefficients.

correlation_flag_threshold:

Defines the minimum acceptable correlation between an individual gait cycle and the average gait cycle for the same person within a recording. The default value is '0.7'. This threshold provides an indicator of data quality for individual gait cycles.

Feature Extraction Workflow

At the beginning of the feature extraction process, the database is cleaned of all features it still holds.

The feature extraction process iterates over all recordings and sensor locations independently. For each combination, the following workflow is executed:

Data Selection and Segmentation

Raw accelerometer data for the current recording and sensor location is loaded and ordered by time. The recording is then segmented according to activity. Some recordings contain a single activity (e.g., continuous walking), others contain transitions between multiple activities. Only segments where the activity ID is included in activities_list are processed further. Each unique activity is processed independent from each other.

Coordinate System Alignment

The accelerometer data is rotated into a unified world coordinate system. This step ensures consistency across recordings, as sensor placement and internal sensor axes vary. See Section 0 for more information.

Resampling

The accelerometer data is resampled to the frequency specified by resample_target_sample_rate.

Jerk Vector Extraction

The magnitude of the jerk (derivative of acceleration) is computed and smoothed using a moving average filter. The filter window is defined by moving_average_filter_window_size.

Fourier Feature Extraction

A predefined number of Fourier coefficients, as specified by fourier_coefficient_amount, is extracted from the jerk signal.

Peak Detection and Gait Cycle Extraction

Dominant and non-dominant peaks as well as minima are identified in the signal. Using these features, individual gait cycles are extracted. Details on peak detection and cycle segmentation can be found in Section *0*.

Normalization and Resampling of Gait Cycles

Each gait cycle is processed as follows:

1. Normalized to the range 0–1, with the 95th percentile used as the maximum value.
2. Resampled to the sample length specified by normalize_target_cycle_length (default: 200 samples).

Write Extrema Events to Database

During the earlier peak detection and gait cycle extraction step, information about (non)-dominant peak and minima position was gained. This information is now saved to the database.

Mean Gait Cycle Computation and Quality Assessment

After extracting and normalizing the individual gait cycles for a segment, the mean gait cycle is computed by averaging all cycles within the current segment. This averaged cycle serves as a reference for evaluating the quality of each individual cycle.

Correlation-Based Quality Measure

For every gait cycle, a correlation coefficient is calculated between the cycle and the segment's mean gait cycle. A cycle is considered usable for further analysis if its correlation coefficient is greater than or equal to the value defined by correlation_flag_threshold. Cycles falling below this threshold are excluded from downstream feature calculations due to insufficient signal quality.

Statistical Feature Calculation

All gait cycles that pass the correlation threshold are used for computing statistical features. The number of cycles included in each statistical calculation is determined by the entries in feature_frames_list. This allows extracting statistical features at multiple scales.

Global Correlation Metrics

Finally, two additional correlation measures are computed for each cycle:

1. Correlation between the cycle and the mean of all cycles from the same subject, across all recordings.
2. Correlation between the cycle and the global mean cycle derived from the entire dataset the cycle belongs to.

These global correlation values provide additional insights into gait consistency both **intra-subject**, by quantifying how closely an individual cycle resembles the subject's typical gait pattern, and **inter-subject**, by assessing similarity to the overall population of one dataset. This allows deviations at both the individual and group level to be identified and supports the evaluation of gait stability and variability across recordings.

Gait Extraction

Since a gait cycle is defined as signal between two consecutive heel strike events of the same leg and heel strike events are located close to the dominant and non-dominant peaks of the signal, the algorithm is essentially a peak detection algorithm.

It consists of two main stages:

- **dominant peak detection**
- **minima and non-dominant peak detection**

These stages are applied sequentially to extract all relevant peaks and segment the signal into discrete gait cycles. The algorithm for non-centered sensors will be shown here in depth with an exemplary sample from the Vajdi Gait Database (Fig 15).

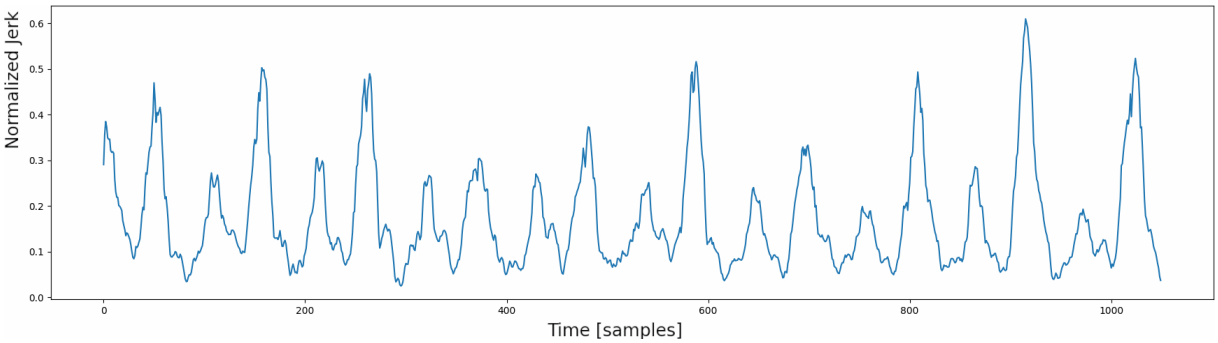


Fig 15. Exemplary signal from the Vajdi Gait Database to show the gait cycle extraction algorithm.

Dominant Peak Detection

Global Step Frequency Estimation

The global step frequency of the signal is estimated via Fourier transformation. The first significant peak in the frequency domain is selected, typically corresponding to the dominant peak frequency. For centered sensor placements, this frequency is approximately 2 Hz; for non-cantered placements, it is closer to 1 Hz.

Peak Identification

- **Centered Sensors**:
  A distance-based peak detection method is applied using the estimated step frequency. Due to uniform peak amplitudes, leg-specific identification is not feasible. Therefore, every second peak is assumed to originate from the same leg, and gait cycles are extracted by segmenting the signal at every second minimum between peaks.

- **Non-Centered Sensors**:
  A mean threshold is computed across the signal. Peaks exceeding this threshold are identified, and a minimum inter-peak distance is enforced to avoid clustering. A second threshold is then applied to isolate **major peaks**, representing dominant heel strikes. Peaks too close to major peaks are excluded to prevent misclassification of high-amplitude non-dominant peaks (Fig 16).

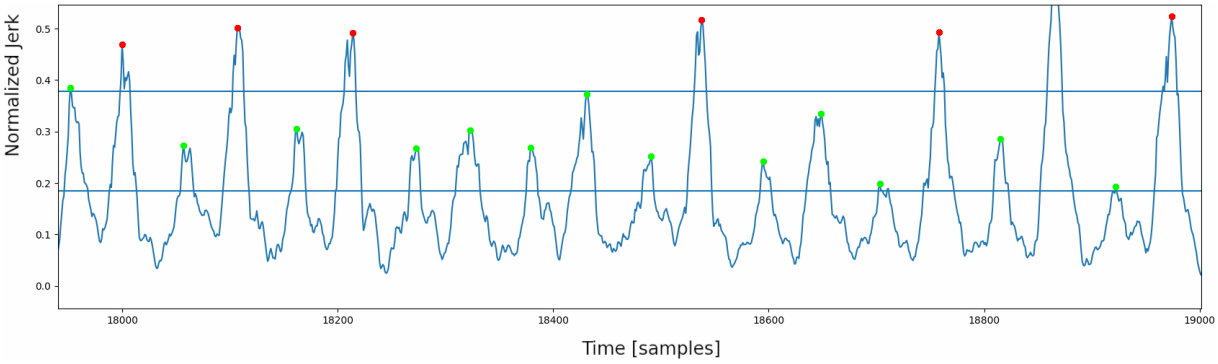


Fig 16. Results of the peak detection.

Green points show peaks above the first threshold, but below the second. Red points show major peaks. It can be observed that there are peaks that are not marked because of minimum inter-peak distance. Note also the five green points in a row, without any major peaks between them.

Gap-Based Peak Interpolation

Gaps between major peaks are analysed to identify missing cycles. The local step frequency is estimated for each gap and compared to its duration to infer the number of missing peaks. Assuming alternating dominant and non-dominant peaks, the missing dominant peaks are interpolated accordingly (Fig 17).

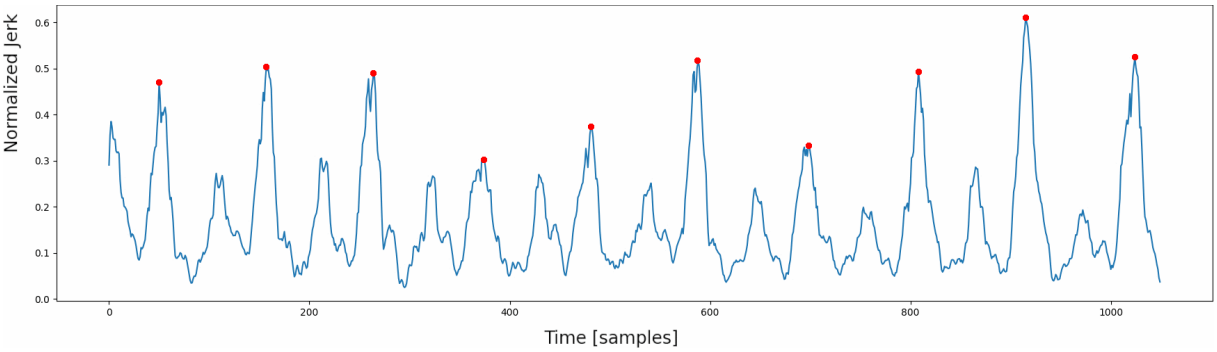


Fig 17. Final result of dominant peak detection for the exemplary signal.

Minima and Non-Dominant Peak Detection

With dominant peaks identified, the signal is segmented into intervals between consecutive dominant peaks (Fig 18). Each segment is expected to contain one non-dominant peak.

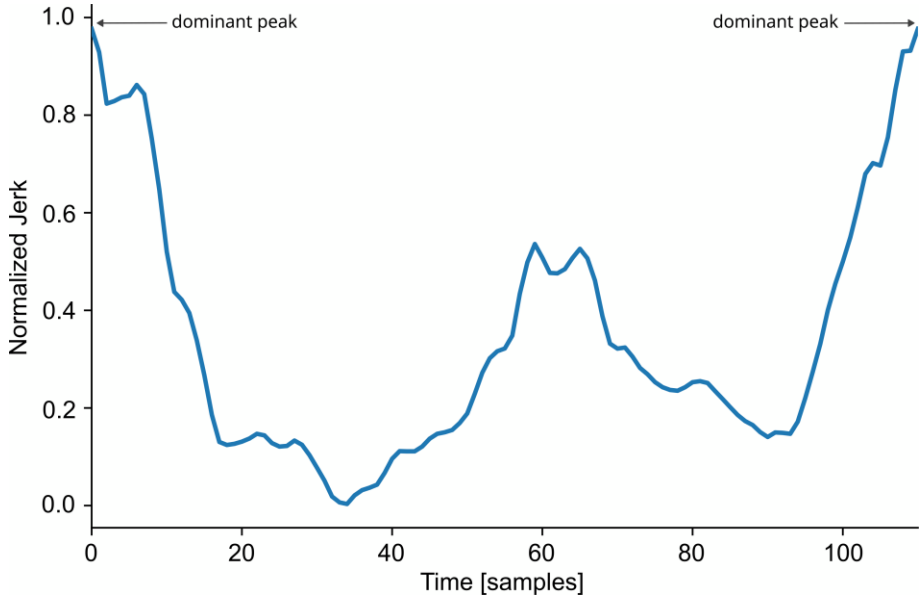


Fig 18. Exemplary segment between two dominant peaks.

Major Minima Identification

Each segment is analysed to identify two **major minima** surrounding the non-dominant peak (Fig 19):

1. **First Major Minimum**:
   The absolute minimum in the segment is selected.
2. **Second Major Minimum**:
   Minima near dominant peaks are excluded. For each remaining minimum, the maximum signal value between it and the absolute minimum is computed. Minima in the same valley as the absolute minimum yield lower values, while those in the opposite valley yield the value of the non-dominant peak. The lowest of these is selected as the second major minimum.

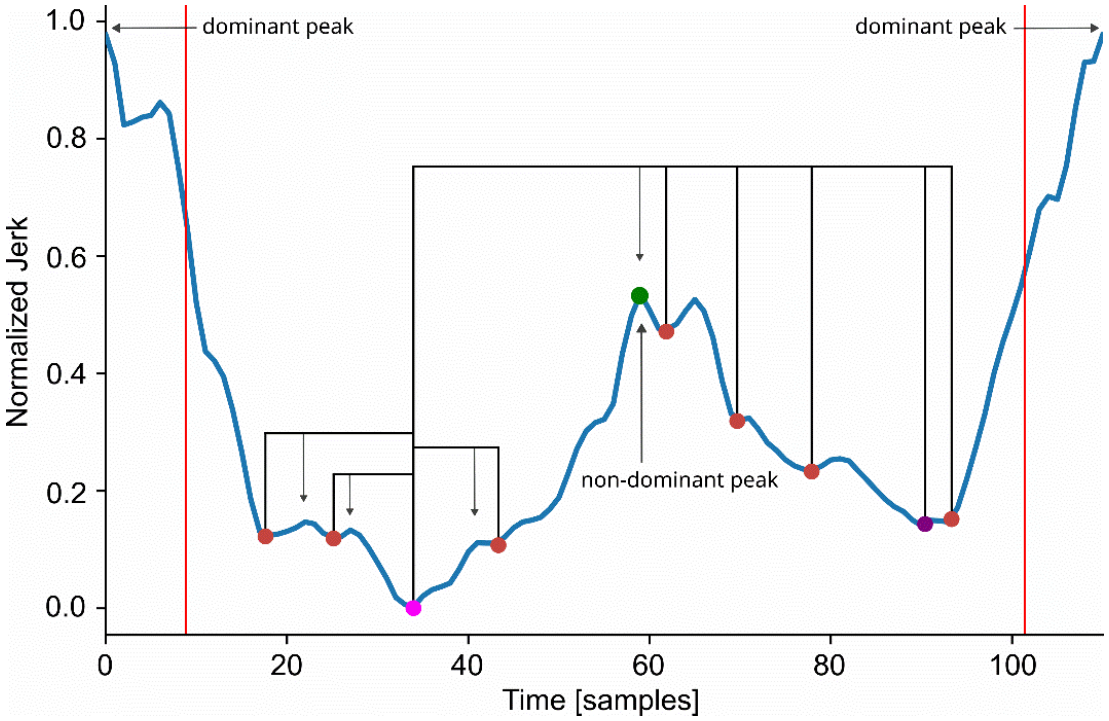


Fig 19. Major Minima Identification.

All detected minima are marked with a circle. The first minimum is simply the lowest value (hot pink). The second minimum is not the second lowest value since this can be on the same side of the non-dominant peak. The algorithm described in the text finds the true second minimum (dark purple).

Non-Dominant Peak Extraction

The non-dominant peak is defined as the highest point between the two major minima (**Fig 20**). This process is repeated for all segments, resulting in a complete set of dominant and non-dominant peaks.

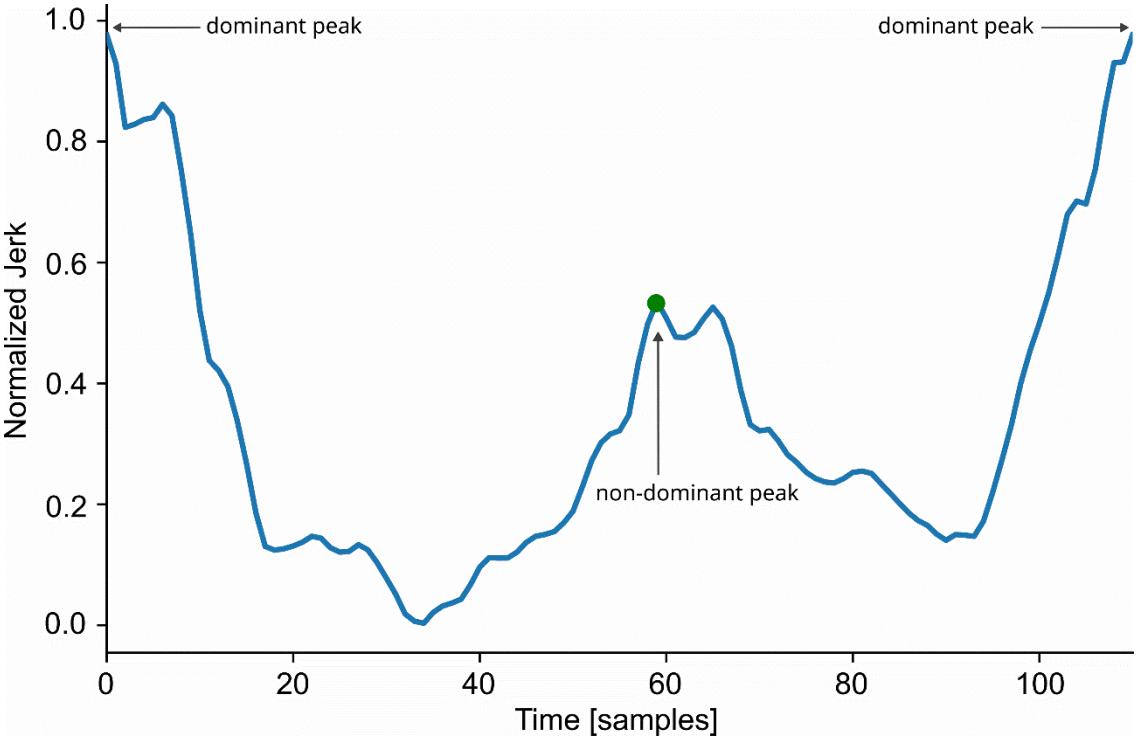


Fig 20. Detected non-dominant peak.

Gait Cycle Extraction

With all relevant peaks identified, gait cycles are extracted by segmenting the signal between consecutive dominant peaks of the same leg. This method ensures accurate cycle boundaries and supports robust feature extraction for subsequent analysis.

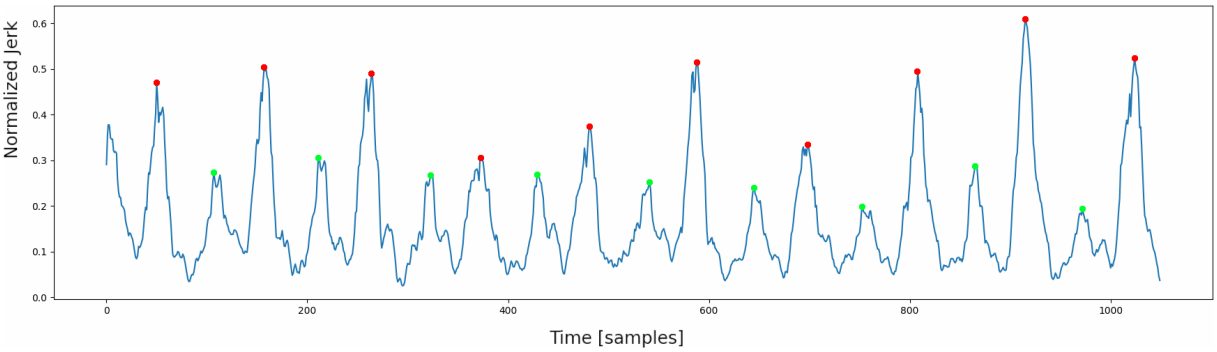


Fig 21. Final result of gait cycle extraction.

Red points represent dominant, green points non-dominant peaks.

References

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